Table 3.4. DC Characteristics (Continued)

Parameter	Test Condition	Min	Typ	Max	Unit
Reset Current					
Quiescent current, nRESET asserted	Typ at 25 °C/3.0 V Max at 85 °C/3.6 V	—	2	3	mA
Processor and Peripheral Currents					
ARM® Cortex™-M3, RAM, and flash memory	25 °C, 1.8 V memory and 1.25 V core ARM® Cortex™-M3 running at 12 MHz from crystal oscillator Radio and all peripherals off	—	7.5	—	mA
ARM® Cortex™-M3, RAM, and flash memory	25 °C, 1.8 V memory and 1.25 V core ARM® Cortex™-M3 running at 24 MHz from crystal oscillator Radio and all peripherals off	—	8.5	—	mA
ARM® Cortex™-M3, RAM, and flash memory sleep current	25 °C, 1.8 V memory and 1.25 V core ARM® Cortex™-M3 sleeping, CPU clock set to 12 MHz from the crystal oscillator Radio and all peripherals off	—	4.0	—	mA
ARM® Cortex™-M3, RAM, and flash memory sleep current	25 °C, 1.8 V memory and 1.25 V core ARM® Cortex™-M3 sleeping, CPU clock set to 6 MHz from the high frequency RC oscillator Radio and all peripherals off	—	2.5	—	mA
Serial controller current	For each controller at maximum data rate	—	0.2	—	mA
General purpose timer current	For each timer at maximum clock rate	—	0.25	—	mA
General purpose ADC current	At maximum sample rate, DMA enabled	—	1.1	—	mA
USB active current			1		mA
USB suspended mode current	1.8 V memory and 1.25 V core ARM® Cortex™-M3 sleeping, CPU clock set to 3 MHz from the high frequency RC oscillator. Radio and all peripherals off			2.5	mA
RX Current					
Radio receiver, MAC, and baseband	ARM® Cortex™-M3 sleeping, CPU clock set to 12 MHz	—	23.5	—	mA
Total RX current (= I_{Radio} receiver, MAC and baseband, CPU + IRAM, and Flash memory)	25 °C, VDD_PADS=3.0 V ARM® Cortex™-M3 running at 12 MHz	—	27.0	—	mA
	25 °C, VDD_PADS=3.0 V ARM® Cortex™-M3 running at 24 MHz	—	28.0	—	mA

Table 3.4. DC Characteristics (Continued)

Parameter	Test Condition	Min	Typ	Max	Unit
Boost mode total RX current (= I_{Radio} receiver, MAC and baseband, CPU+ IRAM, and flash memory)	25 °C, VDD_PADS=3.0 V ARM® Cortex™-M3 running at 12 MHz	—	29.0	—	mA
	25 °C, VDD_PADS=3.0 V ARM® Cortex™-M3 running at 24 MHz	—	30.0	—	mA
TX Current					
Radio transmitter, MAC, and baseband	25 °C and 1.8 V core; max. power out (+3 dBm typical) ARM® Cortex™-M3 sleeping, CPU clock set to 12 MHz	—	27.5	—	mA
Total TX current (= I_{Radio} transmitter, MAC and baseband, CPU + IRAM, and flash memory)	25 °C, VDD_PADS=3.0 V; maximum power setting (+8 dBm); ARM® Cortex™-M3 running at 12 MHz	—	44	—	mA
	25 °C, VDD_PADS=3.0 V; +3 dBm power setting; ARM® Cortex™-M3 running at 12 MHz	—	31.5	—	mA
	25 °C, VDD_PADS=3.0 V; 0 dBm power setting; ARM® Cortex™-M3 running at 12 MHz	—	29	—	mA
	25 °C, VDD_PADS=3.0 V; minimum power setting; ARM® Cortex™-M3 running at 12 MHz	—	24	—	mA
	25 °C, VDD_PADS=3.0 V; maximum power setting (+8 dBm); ARM® Cortex™-M3 running at 24 MHz	—	45	—	mA
	25 °C, VDD_PADS=3.0 V; +3 dBm power setting; ARM® Cortex™-M3 running at 24 MHz	—	33.5	—	mA
	25 °C, VDD_PADS=3.0 V; 0 dBm power setting; ARM® Cortex™-M3 running at 24 MHz	—	30	—	mA
	25 °C, VDD_PADS=3.0 V; minimum power setting; ARM® Cortex™-M3 running at 24 MHz	—	24	—	mA